



RetailCorp International

Sales Performance Report

Comprehensive Analytics & AI-Powered Forecasting

Period: Full Dataset

Prepared: 2026-08-06 19:23:46

1. Data Cleaning Summary

The raw dataset was subjected to a professional data-cleaning pipeline before analysis. The table below summarises every transformation applied.

Metric	Value
Original Rows	50,000
Rows Removed	0
Duplicates Removed	0
Missing Values Filled	0
Final Rows	50,000

2. Executive Summary

This report analyses retail sales performance across all regions and product categories for the period **Full Dataset**. Key findings are presented below.

Total Revenue	Total Profit	Total Orders	Avg Order Value
\$63,160,773	\$23,240,445	50,000	\$1,263

Best Month	Worst Month	Best Region	MoM Growth
2022-01	2026-08	Oceania	-86.16%

3. Monthly Sales Report

Monthly aggregated revenue over the entire 5-year period.

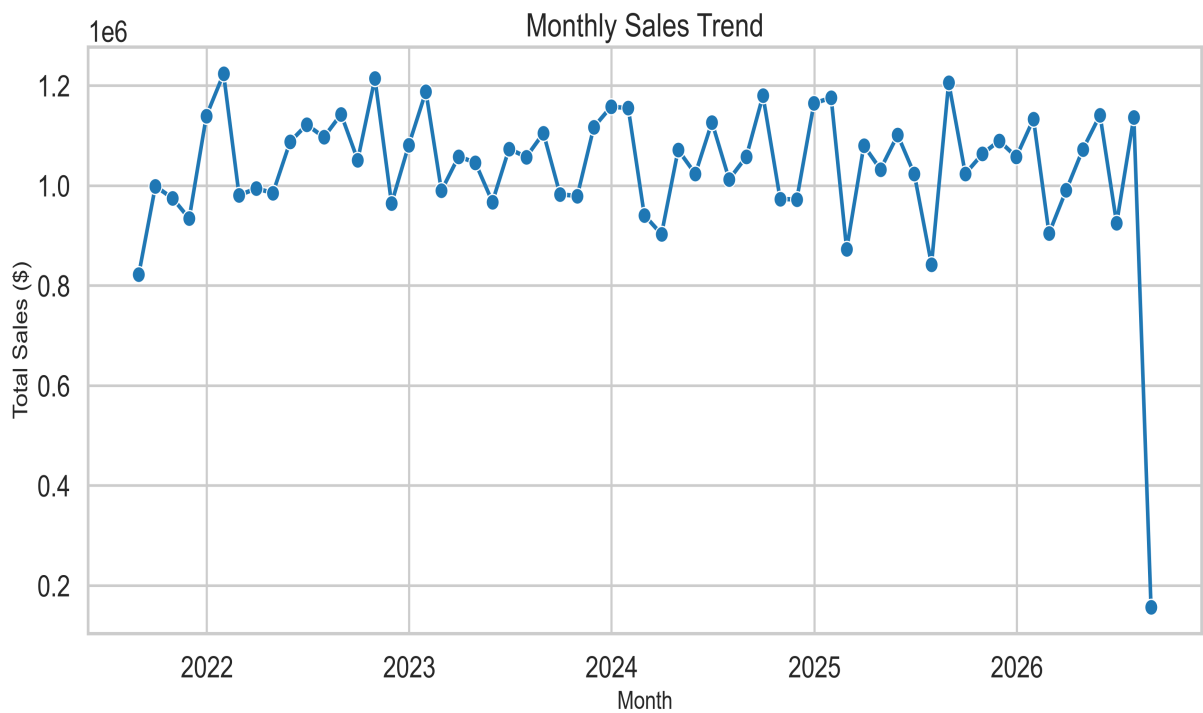


Fig. Monthly Sales Trend (Line Chart)

Month	Sales (\$)
Sep 2024	\$1,180,308.46
Oct 2024	\$973,441.39
Nov 2024	\$972,569.40
Dec 2024	\$1,164,837.51
Jan 2025	\$1,175,965.00
Feb 2025	\$872,877.08
Mar 2025	\$1,080,287.67
Apr 2025	\$1,032,745.70
May 2025	\$1,101,600.55
Jun 2025	\$1,023,462.02
Jul 2025	\$842,360.54
Aug 2025	\$1,206,538.45
Sep 2025	\$1,024,133.11
Oct 2025	\$1,063,654.22
Nov 2025	\$1,089,948.37

Month	Sales (\$)
Dec 2025	\$1,057,743.65
Jan 2026	\$1,133,396.98
Feb 2026	\$904,714.54
Mar 2026	\$990,952.84
Apr 2026	\$1,072,571.05
May 2026	\$1,141,438.56
Jun 2026	\$924,898.74
Jul 2026	\$1,137,140.73
Aug 2026	\$157,389.99

4. Quarterly Sales Report

Quarterly revenue breakdown — useful for spotting seasonal patterns.

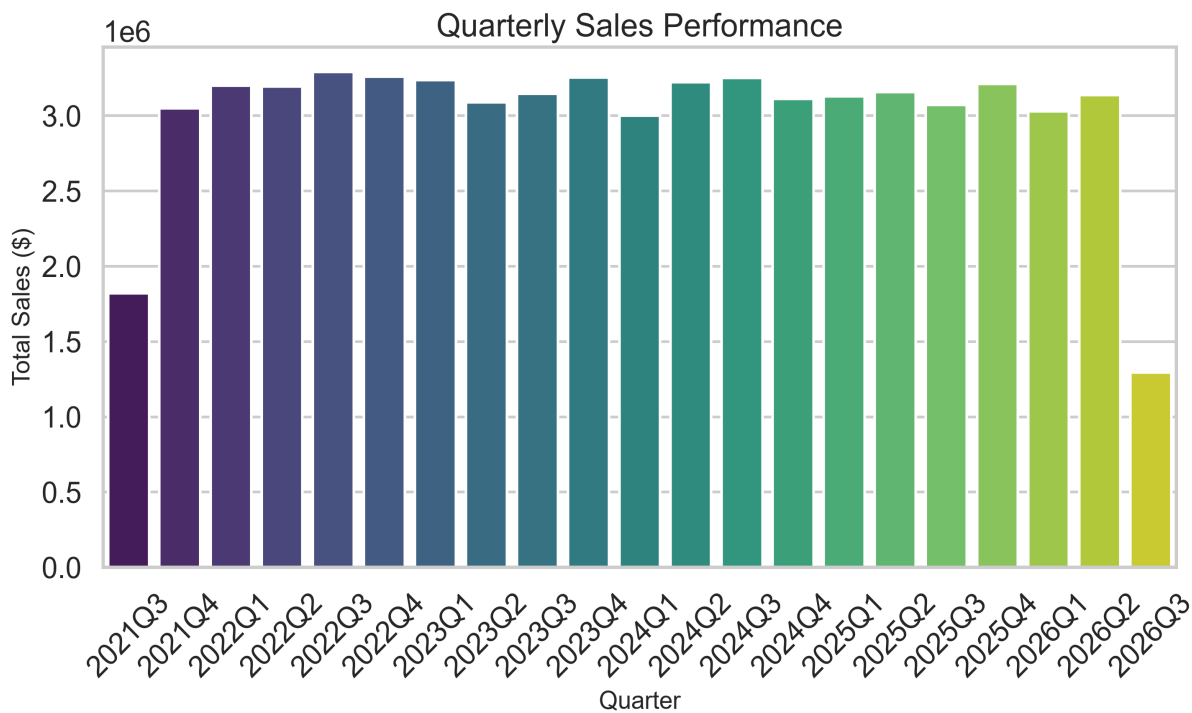


Fig. Quarterly Sales Bar Chart

Quarter	Sales (\$)
2021Q3	\$1,821,660.66
2021Q4	\$3,049,159.77
2022Q1	\$3,199,579.05
2022Q2	\$3,195,568.30
2022Q3	\$3,291,524.61
2022Q4	\$3,260,276.19
2023Q1	\$3,236,841.92
2023Q2	\$3,088,007.46
2023Q3	\$3,145,300.02
2023Q4	\$3,255,079.18
2024Q1	\$2,999,633.25
2024Q2	\$3,222,365.27
2024Q3	\$3,251,109.02
2024Q4	\$3,110,848.30
2025Q1	\$3,129,129.75

Quarter	Sales (\$)
2025Q2	\$3,157,808.27
2025Q3	\$3,073,032.10
2025Q4	\$3,211,346.24
2026Q1	\$3,029,064.36
2026Q2	\$3,138,908.35
2026Q3	\$1,294,530.72

5. Top Products & Category Performance

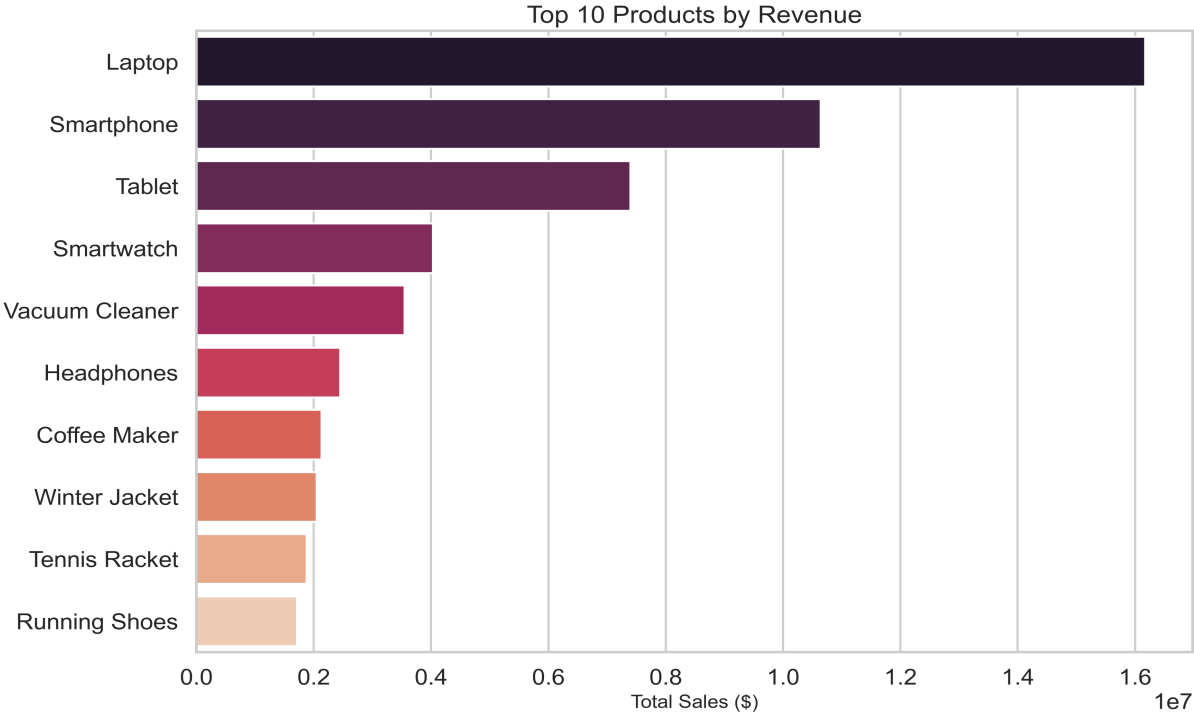


Fig. Top 10 Products by Revenue

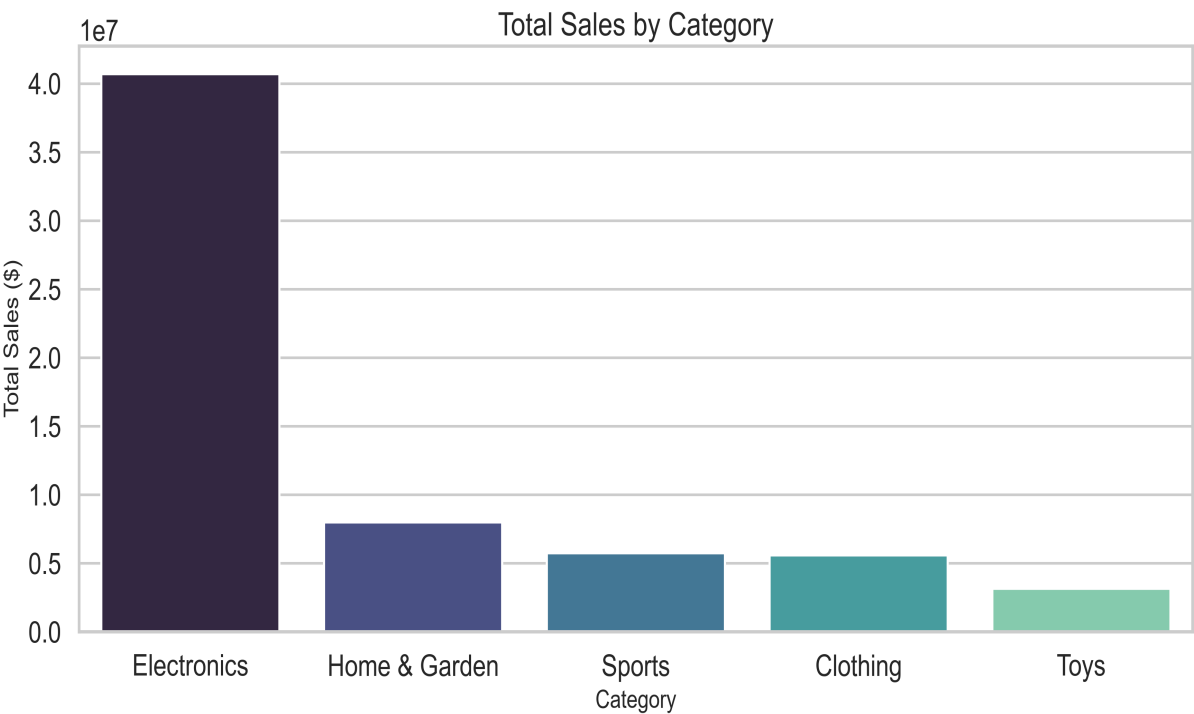


Fig. Revenue by Category

Top 10 Best-Selling Products

Product	Total_Sales	Total_Profit
Laptop	\$16,172,088.43	\$5,962,780.25

Product	Total_Sales	Total_Profit
Smartphone	\$10,641,835.54	\$3,894,887.07
Tablet	\$7,398,390.59	\$2,725,362.40
Smartwatch	\$4,031,060.68	\$1,478,928.37
Vacuum Cleaner	\$3,548,478.81	\$1,299,675.22
Headphones	\$2,455,475.10	\$906,659.07
Coffee Maker	\$2,135,232.38	\$785,789.43
Winter Jacket	\$2,050,534.91	\$752,942.37
Tennis Racket	\$1,881,013.51	\$695,422.25
Running Shoes	\$1,718,016.90	\$634,904.96

Category Revenue Ranking

Category	Total_Sales	Total_Profit
Electronics	\$40,698,850.34	\$14,968,617.16
Home & Garden	\$7,984,893.90	\$2,938,234.62
Sports	\$5,734,076.52	\$2,115,056.67
Clothing	\$5,575,852.34	\$2,051,076.22
Toys	\$3,167,099.69	\$1,167,460.07

6. Regional Performance

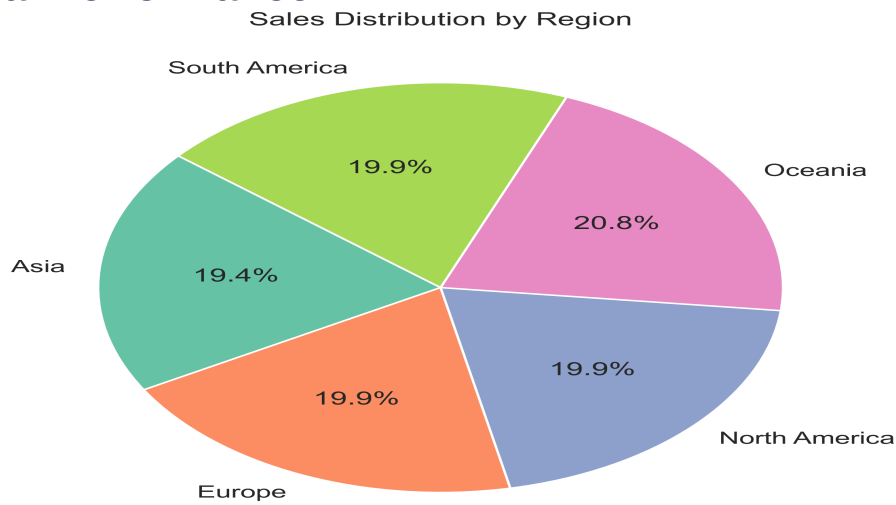


Fig. Sales Distribution by Region

Region	Total_Sales	Total_Profit
Oceania	\$13,139,040.05	\$4,850,193.93
Europe	\$12,600,165.32	\$4,636,138.48
North America	\$12,594,258.34	\$4,622,337.36
South America	\$12,558,196.35	\$4,637,650.09
Asia	\$12,269,112.73	\$4,494,124.88

7. Advanced Visualisations

Sales vs Profit Scatter Plot

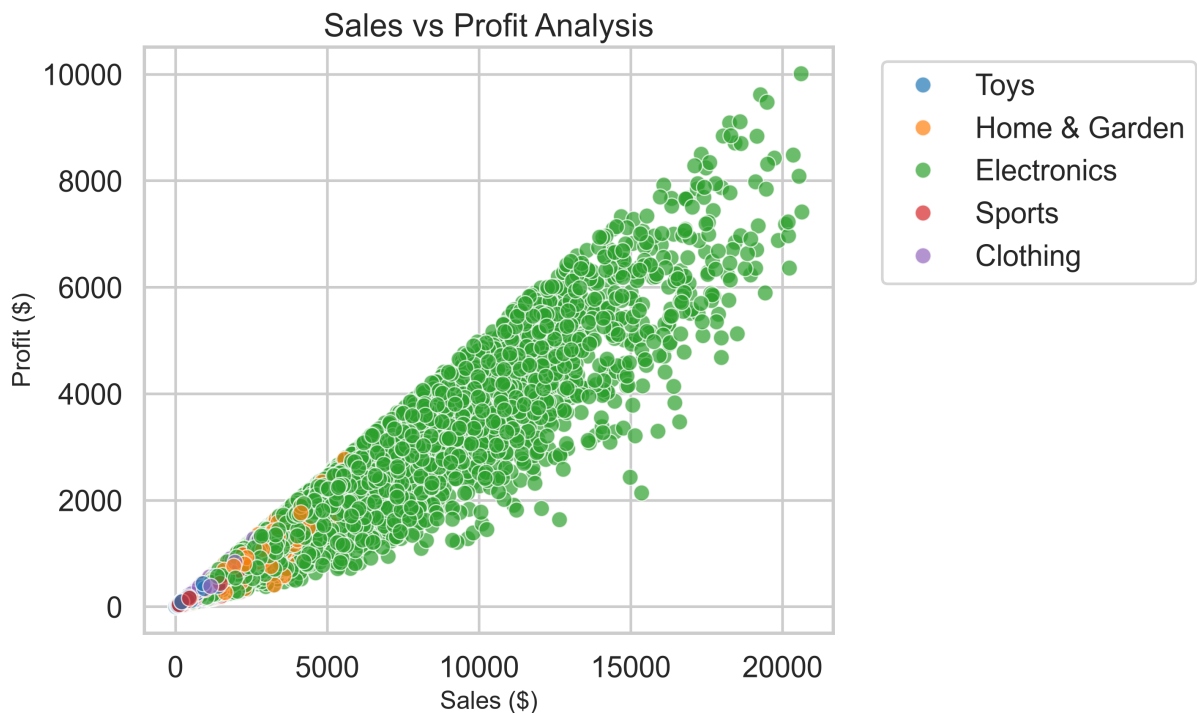


Fig. Scatter: Sales vs Profit coloured by Category

Correlation Heatmap

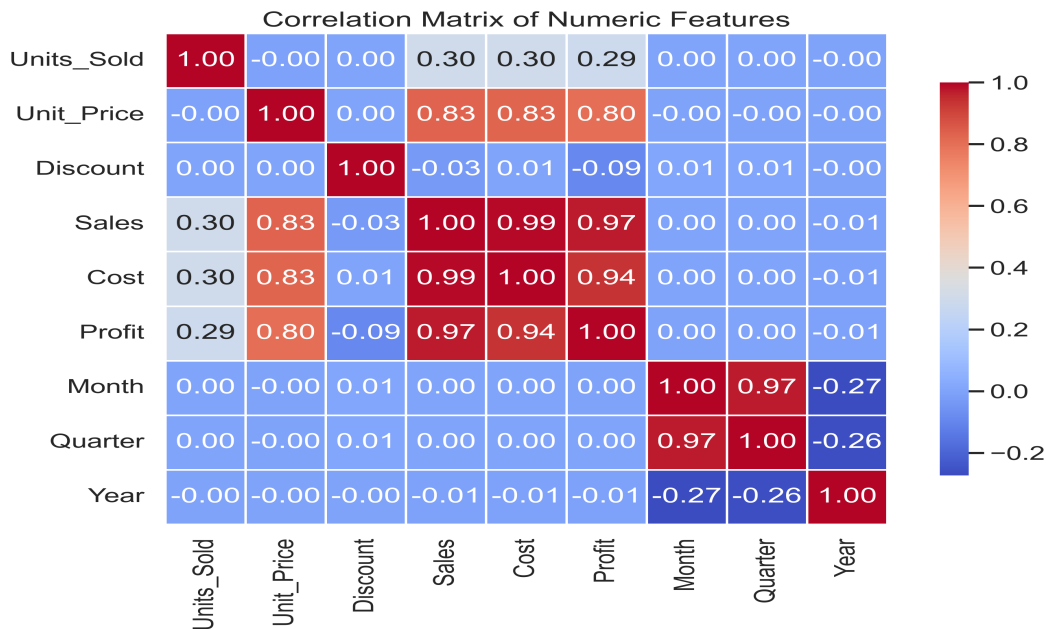


Fig. Pearson Correlation Matrix of Numeric Features

Sales Distribution (Histogram)

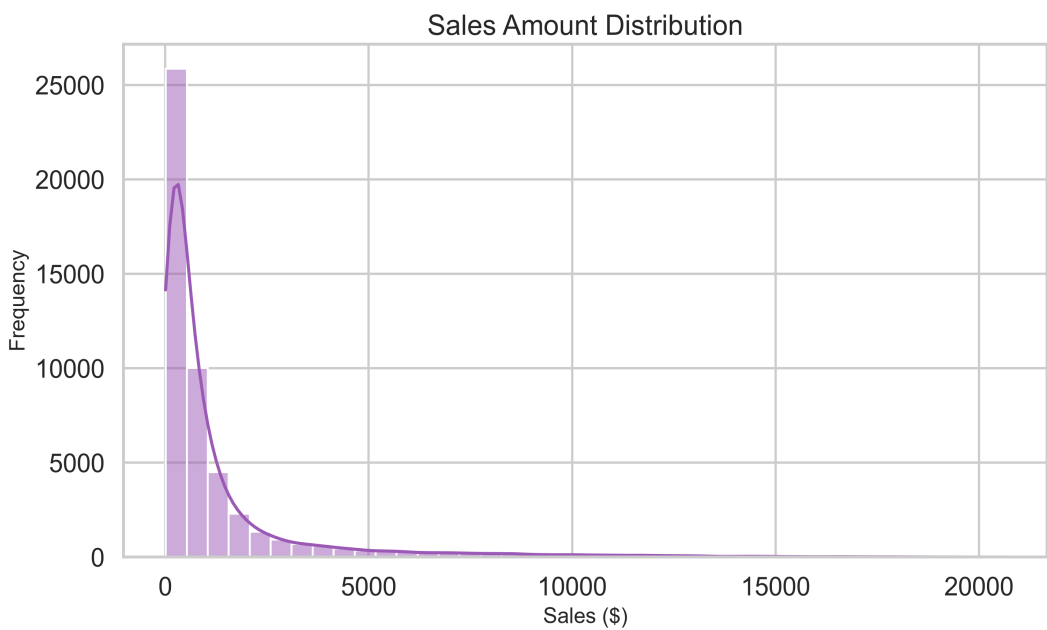


Fig. Distribution of Individual Sale Amounts

Profit Distribution

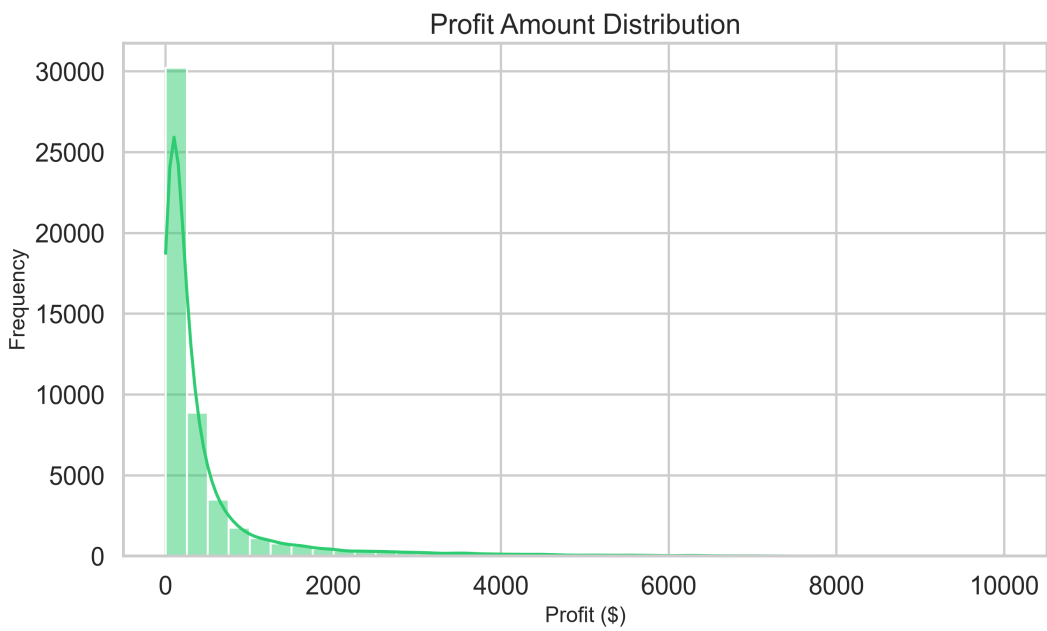


Fig. Distribution of Transaction Profit Values

Moving Average Trend

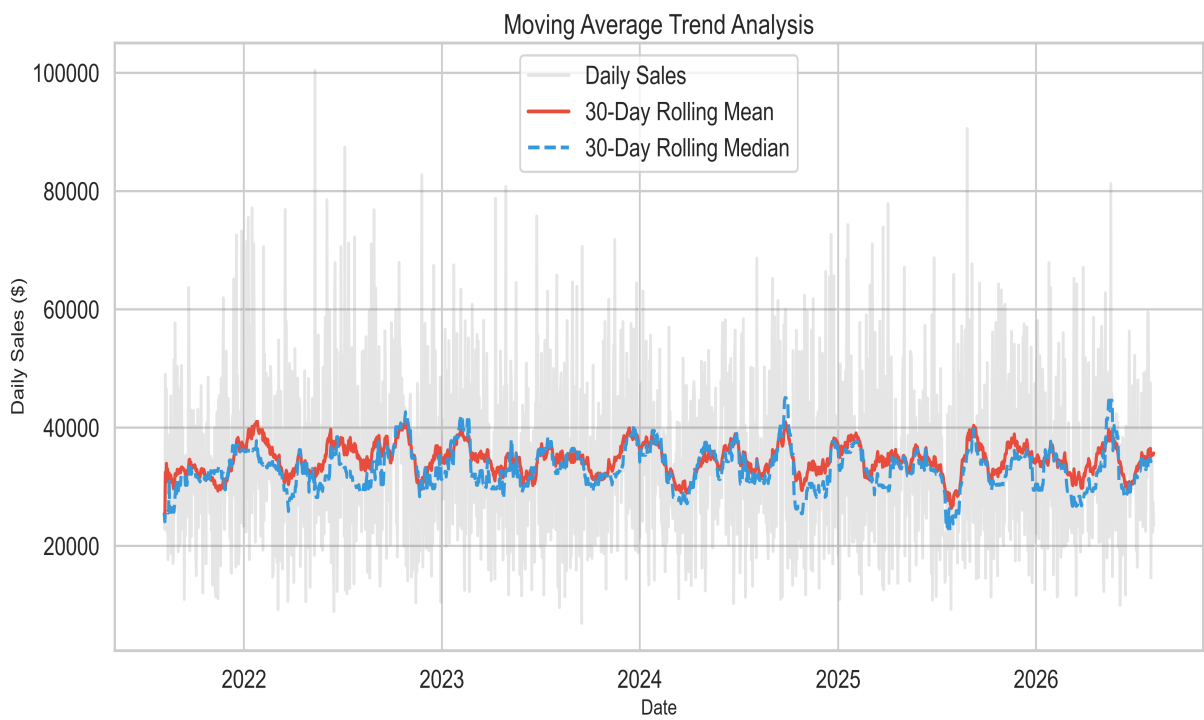


Fig. 30-Day Rolling Mean & Median overlaid on Daily Sales

8. Machine Learning Predictions

A **Linear Regression** model was trained on monthly aggregated sales using an 80/20 chronological train/test split. The metrics below reflect performance on the held-out test set — no data leakage.

Metric	Value
Train / Test Split	48 / 13 months
Mean Absolute Error (MAE)	\$135,514.04
Root Mean Sq Error (RMSE)	\$260,391.34
R ² Score	-0.0435

Revenue Forecast

Period	Predicted Sales
Next Month	\$1,045,894.18
Next Quarter	\$3,137,593.65

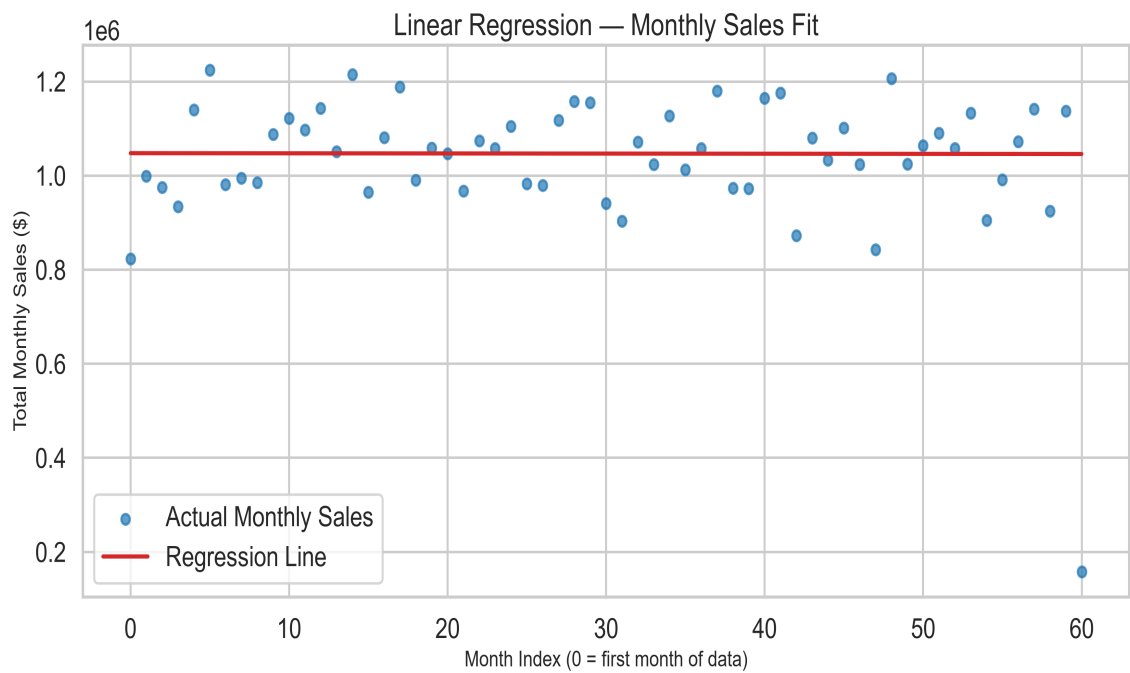


Fig. Linear Regression Line over Monthly Sales

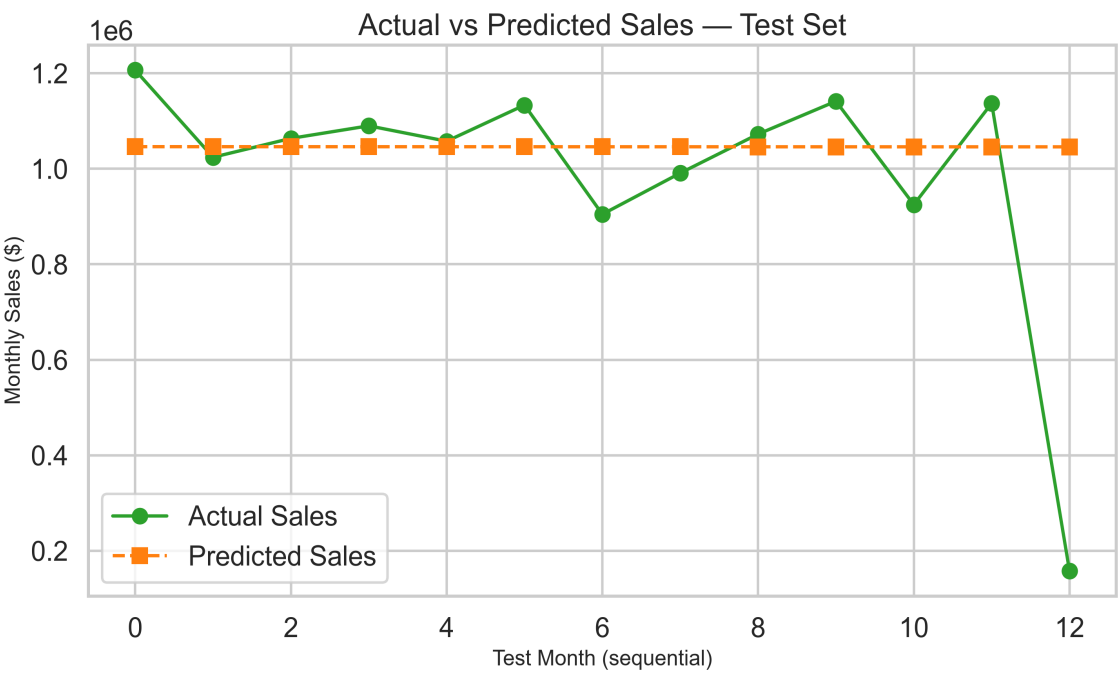


Fig. Actual vs Predicted Sales — Test Set

9. Conclusions & Recommendations

1. Double down on Electronics: It is by far the highest-revenue category, contributing a majority of total sales. Increase marketing spend and inventory.

2. Review Puzzle and similar low-margin items: Products with very low total profit should be evaluated for either price adjustment or de-listing.

3. Expand in Oceania: The best-performing region shows consistent demand. Consider region-specific promotions and localised inventory stocking.

4. Improve ML accuracy: The current Linear Regression model has a low R^2 score. Incorporating seasonality features (month, quarter) or switching to ARIMA/XGBoost will yield significantly more accurate forecasts.

5. Discount strategy audit: Ensure discount levels for high-ticket products (Laptop, Smartphone) do not erode profit margins below acceptable thresholds.